

Emmanouil Giannakis

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EDUCATION

University of Piraeus , Piraeus, Greece	Sept 2021 – Sept 2025
BSc in Digital, Majoring in Computational Infrastructures & Services	
Thesis under the supervision of the Institute of Nuclear and Particle Physics at NCSR Demokritos: "Neutrino's Event Reconstruction with Graph Neural Networks in ANNIE"	Grade: 10/10
Significant Projects:	
- Developed a Sentiment Analysis system utilizing widely adopted NLP techniques and machine learning models, focusing on optimizing accuracy, performance, and computational efficiency for scalable deployment. - Designed and implemented a multi-agent system with dynamic task allocation, negotiation protocols, and performance evaluation in a collaboration-driven environment.	
Lyceum of Ayia Paraskevi , Athens, Greece	2019 – 2021
Honor Roll Student , High School Diploma Grade: 19.0 (GPA 4.0)	
Henry Hudson Elementary School , Vancouver, Canada	Sept 2013 – Jun 2015
Studied at an international school and achieved native-level proficiency in English.	

EXPERIENCE

Head of Autonomous System , Prom Racing FSAE, NTUA – Athens, Greece	Sept 2024 – Sept 2025
Technical:	
- Familiarized with sensor calibration and fusion algorithms, including integration of data from IMU, INS, and camera-LiDAR systems. Gained experience in techniques to improve sensor alignment and accuracy, contributing to robust multi-sensor systems for autonomous navigation and environmental understanding. - Designed and implemented custom datasets with augmentation for object detection and pose estimation tasks. Utilized state-of-the-art architectures and leveraged YOLO-Pose to unify both tasks. Applied knowledge distillation to optimize model performance, achieving a reduction in inference time and enhanced precision with range detection.	
Managerial:	
- Secured a LiDAR sensor sponsorship, achieving a key milestone pursued for over three years. - Oversaw two sub-team recruitments, meticulously selecting the most suitable candidates through a two-step recruitment process and ensuring their integration into the team through technical and safety training.	
Autonomous System Engineer , Prom Racing FSAE, NTUA – Athens, Greece	Oct 2023 – Sept 2024
- Initiated R&D to evaluate the viability of computer vision projects and their investment potential. - Customized a path planning algorithm inspired by RRT (Rapidly Exploring Random Trees) to fit specific requirements, integrating it into the ROS2 framework for enhanced autonomous navigation and decision-making. - Presented the team's perception pipeline at a ZEISS competition, achieving 4th place out of 42 teams (top 10%).	
Student Researcher , NCSR Demokritos – Athens, Greece	Sept 2024 – Present
- Designing and fine-tuning GraphNet-based models for neutrino event reconstruction, leveraging training from similar experiments and analyzing results to guide model improvements (manuscript in preparation for submission to JINST). - Extensive hands-on experience with Linux systems for daily development, including system navigation, shell scripting and troubleshooting. - Co-developed and presented an open-source toolkit for converting ROOT files to SQLite, Parquet, and H5 formats, adopted in collaboration with the Fermilab team.	

SKILLS

Technical Skills: Computer Vision, Digital Image Processing, Machine Learning, Deep Learning, Sensor Fusion, Filtering and Calibration Algorithms, Environment Perception, Multi-Agent Systems, Probability Theory, Stochastic Processes, Model Evaluation, Data Engineering and Visualization, and Testing.

Interpersonal Skills: Teamwork, Communication, Project Management, Presentation, Public Speaking, Problem Solving.

Languages: English (Proficient), Greek (Native).

Programming and Tools: Linux, Git, Bash, ROS2, Python, C, C++, Java, HTML, MATLAB, SQL.

CERTIFICATIONS

Volunteering Assistant , HIAS AI Summer School, NCSR "DEMOKRITOS"	July 2024
Fundamentals of Deep Learning , NVIDIA	May 2024
Unmanned Aerial Systems Common Module , Hellenic Air Force	Apr 2024
Universal AI Course , NCSR "DEMOKRITOS"	July 2023